Shubham Rane

Bachelor of Technology
Electronics and Communication Engineering
Vellore Institute of Technology, Vellore

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., ECE	Vellore Institute of Technology, Vellore	8.54	July, 2027
12th Grade	Central Board of Secondary Education(CBSE)	89.2	May 2023
10th Grade	Indian Certificate of Secondary Education (ICSE)	92.6	July 2021

Projects

- **Development of Neuromorphic Memory Devices Using Metal Oxides** *September 2024 - Present*
  Tools: PLD, Semiconductor Characterization, Electrical Testing, XRD, Circuit Simulation. []
  - Designed and developed neuromorphic memory devices using metal oxide materials, focusing on RRAM for brain-inspired computing.
  - Bridged the gap between traditional and neuromorphic computing by mimicking synaptic functions for energy-efficient memory systems.
  - Worked on device fabrication, material characterization, and modeling to improve memory performance and stability.
  - Utilized Pulsed Laser Deposition (PLD), a rare and niche technique, to enhance material properties for neuromorphic applications.
- **Fire-Fighting Robot Prototype** *January 2025 – April 2025*
  Tools: IR sensors, Motor Driver, DC and Servo Motors, Water Pump, Relay module, Arduino Uno
  - Developed an autonomous fire-fighting robot using IR sensors for fire detection, DC motors with L298N driver for navigation, and servo-controlled water nozzle for targeted suppression.
  - Implemented control logic in Arduino IDE, integrating a relay-based water pump system along with LED/buzzer alerts for real-time status indication.
  - Designed an efficient power management system using Li-ion batteries and regulators, with scalability considerations for future IoT and advanced detection integration.

Skills

- **Programming Languages**: Javascript, HTML, CSS, Java, Python, Verilog , R
- **Technologies**: Digital and Analog Circuit Design, Circuit Simulation, Fabrication and Characterization of Nanomaterials, ASIC Design flow
- **Tools**: NI MultiSim, ModelSim, MATLAB, R Studio, Pulsed Laser Deposition, LabView, Cadence Virtuoso

Experience

- **Internship at Team Lambda, VIT Vellore (July to Dec 2025)**: Worked on research-oriented tasks aligned with the team's requirements, including:

  Researching competitions relevant to the team

  Gathering information related to physical components and resources needed for projects

  Supporting decision-making through structured and requirement-based research

Certifications

- **Vellore Institute of Technology, Vellore**, Workshop on EMI/EMC Measurement *February 2025*
- **Vellore Institute of Technology, Vellore**, Value added program on LabView *April 2025*
- **Maven Silicon**, VLSI Physical Design Internship *July 2025*
- **Team Lambda,VIT Vellore**, Research and Marketing Internship *Dec 2025*